Technical report — Operator instructions for lightpath provisioning with Apertus 8B

Every number is copied from the results file named next to it. The long account of the first two rounds is in `docs/report_rounds_1_2.md`; every result with its source file is in `docs/results_overview.md`.

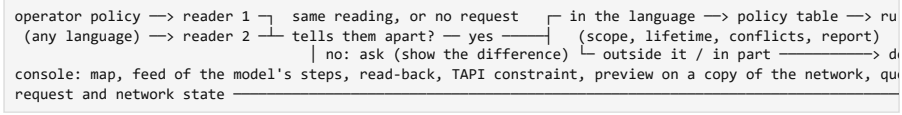
- **Track:** Track 2B, own project: operator instructions for lightpath provisioning
- **Event:** Online
- **Team:** ApertusNetworkGym (Wadood Ranjha)
- **Demo:** TODO link to the video

1. Summary

Operators of optical transport networks state policy in plain language ("the Zurich–Milan span is under maintenance", "for 400G services, avoid Basel until further notice"), in the language they work in. We built a blueprint in which Apertus 8B reads such a policy while the network still follows fixed, checkable rules. Apertus translates the sentence once into one line of a small policy language; a rule engine enforces that line for every request the policy covers, for as long as it holds. A second reader translates it again, and the operator is asked when a request would be decided differently under the two readings. An instruction outside the language goes to a fine-tuned Apertus that decides the request itself, with its choice checked against the part the language carries. The line is also a TAPI 2.1.3 constraint, the object a transport SDN controller takes. An operator console shows the network on a map with a feed of what the model does: it reads back every policy, previews its effect on a copy of the network before it is applied, names conflicts with the policies in force, pauses the stream to ask when two readings would decide a request differently, and keeps an audit trail per lightpath. Everything runs on a laptop GPU.

Measured in the Optical Networking Gym simulator on sentences written by separate AI instances in four languages: the reader is right on 94.8 % of 2,078 instructions, on clause pairs it never saw together as on trained pairs (96.5 against 95.6 %; triples three points lower); its decisions comply in 99.6 %, those of the model that decides alone in 91.1 % (English probes). With scope and lifetime added, a reader trained on the same recipe is right on 93.5 % of 800 blind policy sentences, and in closed loop the policy table built from its readings breaks 1.1–1.5 % of requests against the true table, where ignoring scope breaks 13–30 %, ignoring lifetime 6–12 %, first fit 38–42 %. On a mixed stream the hybrid that asks breaks 8.5 % of requests (decider alone 16.8 %, first fit 28.6 %); asking only when a request tells the readings apart puts 16 % fewer questions at the same error for the job translator, and a third fewer for 0.35 points on the laptop hybrid. The weak points are measured too: the reader silently applies 47 % of instruction types no component was ever shown, and the phrase "to Prague" is a scope to the new reader and a destination to the operator. The 4-bit laptop build, the one a judge can run, is one to four points weaker on every measure and is reported beside the job figures throughout.

2. Architecture



- **Simulator** (`src/sim.py`): Optical Networking Gym, preset `jocn_benchmark` (nobel-eu, 28 nodes, 41 links, five candidate paths per request, 320 slots, six modulation formats); a Swiss topology of 24 cities and 33 links (`data/topologies/swiss.xml`, built from public geography) loads through the same code. Each request is a **request view** (`src/view.py`).
- **Policy language** (`src/spec.py`): `avoid=Berlin-Prague,Paris; via=Lyon; hops=4; km=any; path=any; prefer=km,load; scope=rate:400; until=6h; rest=none`. Links and nodes to avoid, a node to pass through, limits, a candidate by number, preferences in order; **scope** (every request, a bit rate, requests from or to a node) and **until** (hours, or until withdrawn) make a sentence a standing policy. `rest=more` marks a part not carried; `unsupported` and `no_instruction` are the refusals; a name that is not a node makes the line *unresolved*.
- **Readers** (`src/translate.py`): Apertus 8B with a LoRA adapter, decoding under a grammar built from the node list (`src/grammar.py`); the same grammar as GBNF for llama.cpp.
- **Policy table** (`src/standing.py`): which policies are active, which cover a request, the one line they combine to (constraints add up, the tighter limit wins, the newest preference decides, older ones break ties), the conflicts a new policy raises (a node one policy requires and another avoids, two candidates by number, a node the avoided links cut off; an override of a preference), and a report per policy (requests covered, decisions changed, rejected).
- **Rule engine**: constraints filter the candidates (none left: reject), preferences apply in order. With a correct line it cannot break the policy.
- **Decider** (`src/policies.py`): Apertus 8B with another adapter answers `path=1 slot=120` or `reject`. **Guard:** a path that breaks the constraints the line carries is replaced.
- **TAPI** (`src/tapi.py`): the line as a `topology-constraint` and `routing-constraint` of TAPI 2.1.3 (`exclude-link`, `exclude-node`, `include-node`, `max-allowed-hops`, `max-allowed-cost`, `route-objective-function`); what the standard cannot carry travels in `name` entries.
- **Console** (`src/console.py`, `make console`): one stream under standing policies and per-request instructions; a feed that narrates each step of the model (reading, agreement or question, decision, guard); the policy table with conflicts and a preview of each policy on a copy of the simulator; questions put by consequence with both routes; every decision judged against what was meant; the audit trail. From recorded answers, or live behind llama.cpp.

Target architecture (mandatory)

a) On-premise, and it runs **b) air-gapped**. At runtime the system is the network controller (here the simulator), the policy table and one model server; nothing calls out. The server holds one base model quantised to 4 bits (5.1 GB) and four adapters of 160 MB (two readers of the third format, one of the fourth, the decider), switched per request. On a laptop with an RTX 4060 (8 GB) under llama.cpp: 0.89 s per instruction, 0.97 s per decision (`results/design/readers3.json`). A reader is called once per policy, not per request; the decider only for instructions outside the language. The container built by `make run` needs no network. A note on the host: with

the adapters loaded llama.cpp held 7.3 GB of host RAM, and a WSL VM of 7.8 GB left little for the simulator; the laptop runs were slowed by that, not by the GPU.

3. Use of Apertus

- **Model:** `andreasmartin/apertus-v1.5-8b-text`, the text-only conversion of `swiss-ai/Apertus-v1.5-8B`; the organisers accepted it on Discord (7 Oct 2026) provided the fine-tuned weights are published and the procedure explained.
- **Four adapters** (QLoRA: 4-bit NF4 base, r = 32, α = 32, 79.7 M trainable parameters, completion-only loss; `src/train.py`): the decider on 12,000 simulator decisions (138 min); two readers of the third format on 5,570 instructions each (34 min each); one reader of the fourth format on 7,170 (65 min). TODO: public links.
- **Decoding:** greedy, constrained to the decision line or the policy language.
- **Serving:** Transformers in Hugging Face Jobs (one A100) for training and scoring; llama.cpp with our own 4-bit GGUF of the base and GGUF adapters on the laptop; the rendered prompts are identical apart from the start token (`results/train/rendered_prompt3.txt`).
- **Apertus 70B** (both generations, 4-bit GGUF, 15 examples, no training) translated the same test instructions under the same grammar, and generation 1.5 served as the second reader.
- **Other models:** none is part of the system, and none produced a result. The brief allows other open-weights models to support development; we also used a closed one, Claude (Anthropic), and state its role exactly. It wrote code and documentation under the author's direction, the sentence templates of the readers' training data, and, as separate instances that never saw the repository or each other's output, the blind test sentences. It is not used at runtime, it never labelled a training example (every training line comes from the project's generators, which fill templates from the topology and know the line by construction), and it never judged an answer (scoring is exact comparison of lines; the simulator judges decisions). What rests on it is the wording of the sentences, and that is checked by the author's own 22 lines and by the sentences from other people we invite before the demo.

4. Data

Synthetic, no personal data; licence CDLA-Permissive-2.0; the test sets with every reader's answers are exported as a dataset with a card (`src/benchmark_export.py`, `data/benchmark/`: 4,712 instructions, 20 sets).

Data	Size	How it was made
Decider, training	12,000 decisions	simulator episodes (seeds 101–106) labelled by rule-based teachers
Readers, training	5,570 (third format), 7,170 (fourth)	templates in four languages filled from the topology; the fourth adds scope and lifetime clauses
Readers, test	2,078 + 661	168 sentences per language by four separate AI instances from a brief naming the clause kinds and none of our wording
Standing policies, test	800	200 scope and lifetime phrases by another separate instance, attached to the test sentences
Swiss network, test	1,151	the test sentences filled with Swiss cities in each language's own name (*Genf*, *Genève*, *Ginevra*)
The author's lines	22	typed by the owner: 17 English, 5 Urdu in Latin script
Closed loop	2 loads × 3 seeds × 2,000 requests	traffic seeds 1–3, never trained on; 4,599 distinct instructions; a second stream of standing policies

Which clause combinations are trained (eight pairs) and which held out (seven pairs, five triples), and four instruction types absent from all training, were fixed and committed before any test sentence was opened (`7ab1e66`). A request is used for a test instruction when the instruction is *restrictive* there: at least one feasible path complies and one does not.

5. Evaluation

Protocol. A decision *complies* when it lies in the set of paths the instruction allows; with none allowed, only a reject complies. Closed loop: the no-intent teacher fills the network for 1,000 requests, the policy decides 2,000; loads 240 and 390 Erlang, three seeds. Two instruction-blind choices are scored everywhere: the first candidate that fits, and the decision without an instruction. Tables: `results/tables/phase3.md`.

Translate or decide (`results/offline/*_probes3.json`, 672 English requests).

	One clause	Pair seen	Pair never seen	Three clauses, never seen	All
Reader 1, right meaning, %	94.6	95.6	96.5	91.2	94.8
Apertus 8B, 15 examples, no training, %	89.5	81.0	91.1	85.6	87.2
Comply: first candidate that fits, %	70.6	63.3	62.5	57.5	66.8
Comply: decider alone, %	96.2	90.6	85.7	76.2	91.1
Comply: reader 1 and rule engine, %	99.4	100.0	99.1	100.0	99.6

The slots carry the composition; the decider loses ground with every clause, part of which is training rather than architecture: it was taught the first round's four intents, not the third round's clause kinds (its violations split by kind are in `docs/results_overview.md`). By language the reader is right on 96.7 (en), 91.4 (de), 96.0 (fr), 94.8 (it) %. Apertus 70B with examples and no training reaches 89.2 % (generation 1.5; the first 71.9 %): the fine-tuned 8B reads better than a model nine times its size, as Qiu et al. (2022) found for compositional parsing.

The gate and the second reader (`results/design/readers3.json`; "left" = applied wrongly although two readers were compared; in brackets the share asked).

	Reader 1 alone	Left with reader 2	Left with the untrained 8B	Left with the untrained 70B
Policy language, applied wrongly %	4.6	3.7 (2.8)	1.5 (12.7)	0.2 (13.4)
Types never shown to any component (253 sentences, hard and impossible variants included)	46.6	16.2 (31.6)	19.4 (44.7)	0.0 (69.2)
Carried in part	72.2	36.3 (50.9)	33.0 (54.2)	33.0 (53.3)

A learned gate does not recognise what it was never shown. Two readers of one recipe share most misreadings; a reader that differs catches more, which is what Hamidieh et al. (2026) report for cross-model disagreement. The fourth-format reader (section on standing policies), trained on the same recipe with 1,600 more sentences that carry an extra clause, silently applies only 16.7 % of the 108 never-shown sentences (reader 1: 46.3 %) and, as the check behind reader 1, leaves 10.2 % of them at 38.0 % asked; inside the language, on the four-language pool, it leaves 2.0 % at 4.9 % asked where reader 2 leaves 3.7 % at 2.8 %: fewer silent errors for more questions; one training run, so a lead, not a result. A third signal, five readings of reader 1 at temperature 0.8 (`results/design/consistency3.json`; all three signals measured on the laptop's 4-bit readers, the 70B from its job cache): asking when any sample differs from the greedy reading leaves 3.5 % of the policy-language sentences wrong and unasked at 8.0 % asked (reader 2 in 4 bits: 4.2 % at 5.4 %; the 70B: 0.6 % at 15.1 %), and 2.8 % of the never-shown types at 88.0 % asked (reader 2: 8.3 % at 37.0 %). Disagreement among samples marks unfamiliar input more than it marks error, as Cole et al. (2023) and Hamidieh et al. (2026) report, and costs five more readings, each about one point worse than greedy (92.1 % against 93.1 % in the language); reader 2 or any differing sample together leave 2.9 % at 10.1 % asked.

Closed loop, mixed stream (45 % in the language, 15 % unknown types, 10 % in part, 10 % no instruction, 5 % places outside, 15 % none; `results/eval/*_mix3.json`), violations % of requests at 390 Erlang, three seeds: first fit 28.6, decider alone 16.8, translator alone 10.6, hybrid with guard 9.5, hybrid that asks 8.5 (asks for 8.9 % of requests), with the untrained 70B as the check 8.3 (asks 24.1 %). Inside the language the hybrid breaks 0.8 %, on unknown types 28.9 %, and a better gate does not move the latter (28.3 %): the limit is what the decider can follow. Figure 1 puts the two closed loops side by side.

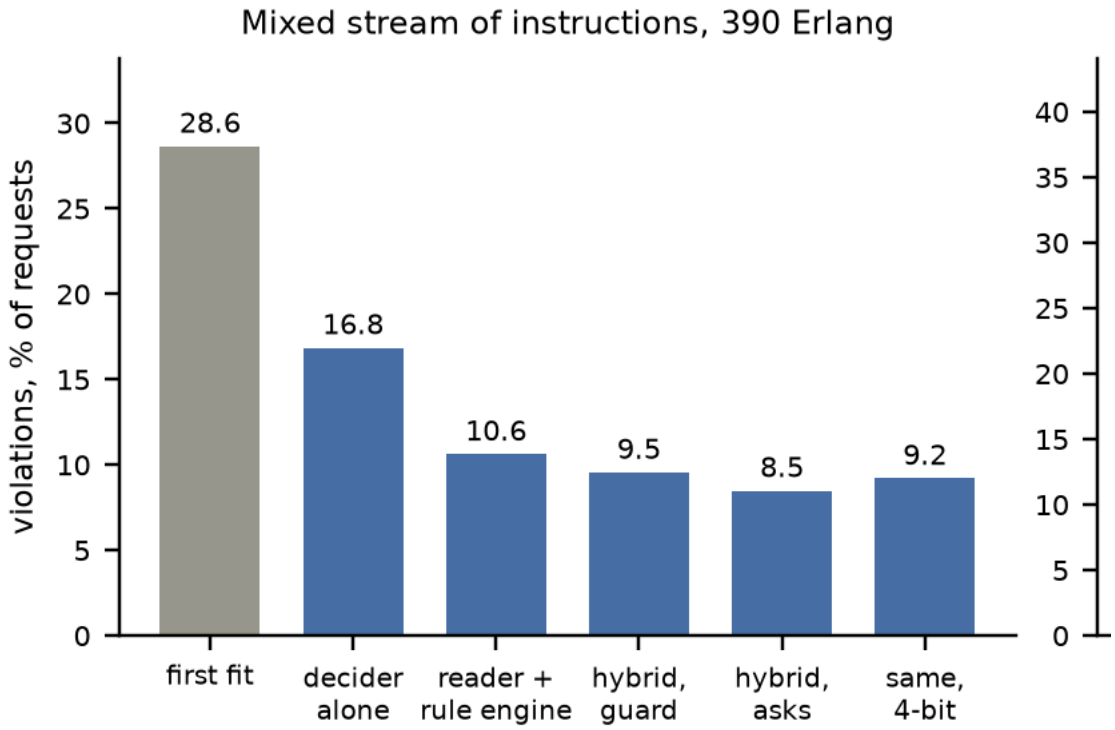


Figure 1. Violations in closed loop, three seeds: the mixed stream of instructions at 390 Erlang (left) and the standing-policy stream judged against the true table (right); `results/eval/`. **Asking by consequence** (`translate3_ft_askif-ft2`, both loads, one process): the translator that always asks puts 397 questions and asks on 8.9 % of requests at 390 Erlang for 10.28 % violations; asking on the first request that tells the readings apart puts 332 (65 pairs of readings were never told apart) and asks on 7.8 % for 10.25 %. On the laptop, with the 4-bit models and the decider live, the hybrid that asks breaks 9.18 % (job: 8.47 %) with 225 questions; asking by consequence breaks 9.53 % with 152 questions (73 pairs never told apart), asking on 7.4 % of requests instead of 9.6 %. The small cost has a cause: when both readers are wrong in different ways, asking at once fixes the reading, asking by consequence only once the two wrong readings diverge. The simulated operator answers with the truth, which an operator shown two wrong readings cannot; with a **fair operator** who picks the right reading only when one of the two is right and otherwise leaves the first (`translate3_ft_askfair-ft2`), the translator breaks 10.3 / 10.7 % at 240 / 390 Erlang against 10.1 / 10.3 % with the truth and 10.5 / 10.6 % without questions: inside the language the fair operator matches the truth (0.8 %), the rest of the gain is the oracle's; of the 397 questions, 167 showed two wrong readings. The hybrid's gain from asking (9.5 to 8.5 %) was measured with the truth only and is an upper bound.

Standing policies (`results/eval/*_standing4.json`). The fourth-format reader on the 800 blind policy sentences: right meaning, scope and lifetime included, 95.0 (en), 92.0 (de), 95.0 (fr), 92.0 (it) %; by kind, a bit rate 90–97.5, requests from a node 72.5–90 (the weakest), to a node 92.5–100, a number of hours 90–100, until further notice 92.5–100. On the third round's sets it is at least as good as reader 1 (96.7 / 93.2 / 96.0 / 94.6 %). In closed loop, policies arrive (three at the start, one every two simulated hours, at most four active), hold for the hours they state or until withdrawn, and cover the requests their scope names; every decision is judged against the true table.

Policy, violations % of requests	240 Erlang	390 Erlang	Rejected, %
The true policy table	0.0	0.0	31.4 / 34.7
First fit (ignores the policies)	42.0	38.1	2.7 / 8.2
True policies, scope ignored	12.7	30.0	43.5 / 61.4
True policies, lifetime ignored	11.9	6.4	42.7 / 39.8
Policies read by the fourth-format reader	1.1	1.5	30.7 / 34.2

Policy, violations % of requests	240 Erlang	390 Erlang	Rejected, %
The same reader in 4 bits on the laptop	2.4	1.6	31.8 / 34.2
Policies read by the third-format reader (no scope, no lifetime)	21.4	36.9	52.0 / 68.1

The stream's policies are restrictive by construction (a third of requests are rejected under the true table); what the ablations show is the cost of ignoring scope or lifetime, and the reader's table stays within 1.5 points of the truth (per seed 0.9 / 0.0 / 2.4 and 0.0 / 4.4 / 0.1). The ablations swing with the seed, the reader's table does not: over ten seeds (`results/eval/*_standing4_seeds10.json`) first fit breaks 41.9 ± 6.3 / 35.7 ± 4.6 %, scope ignored 23.2 ± 13.2 / 23.9 ± 18.0 %, lifetime ignored 19.5 ± 9.1 / 9.2 ± 5.2 %, the reader's table 1.0 ± 1.1 / 2.0 ± 2.1 % (worst seed 5.7 %), the true table 0.0 / 0.01 % (two needless rejects, the cost of its own bookkeeping). A reader of the old format, which cannot read either slot, applies every policy to every request for ever and rejects half to two thirds of them. In 4 bits on the laptop the fourth-format reader is right on 90.6 % of the policy sentences (93.5 % in the job) and its table stays within 2.4 points of the truth.

A Swiss network and the author's lines. On 1,151 sentences with Swiss cities in each language's own name (*Genf*, *Genève*, *Ginevra*, *Sankt Gallen*; no reader ever saw them), reader 1 on the laptop is right on 90.6 (en), 92.0 (de), 90.8 (fr) and 89.0 (it) %, against 93.1 % on the network it was trained on; one clause 92–97 %, three clauses 70–85 % (`results/translate/translate3_ft_laptop_swiss.json`). In closed loop on that network at 600 Erlang (three seeds, `results/eval/*_mix3_swiss.json`; the laptop readers throughout) the translator alone breaks 9.8 % of requests (2.2 % inside the language), the translator that asks 9.0 % (1.2 % inside the language; asked on 12.6 % of requests), asking by consequence 9.3 % (asked on 8.0 %), first fit 24.7 %. The same laptop readers on the European network at 390 Erlang (`results/eval/translate3_ft_laptop*_mix3.json`) break 1.9 % inside the language alone and 1.2 % asking: the unseen city names cost 0.3 points alone and nothing once a second reader is asked. Of the author's 22 lines, reader 1 is right on 18 (16 of 17 in English, 2 of 5 in Urdu), the fourth-format reader on 17: its two new errors read "go from London to Lyon via Oslo" and "to Prague, avoid Berlin" as scopes (`to:Lyon`, `to:Prague`), where the author meant the current request's end points. The language has no word for a request's own end points; that is the next slot to add.

6. Limitations

- One topology in closed loop (the Swiss one offline so far), one traffic model, one simulator; the model ranks candidates the simulator has already checked, it does not compute physics.
- Every test sentence but 22 was written by an AI instance; four languages, no Swiss German in the demo (72.7 % on a stress set).
- Unknown types remain the weak point (28–37 % violations for every policy in the mixed stream).
- The fourth-format reader has no second reader yet, so scope and lifetime are read once; the stream's policies are our construction and reuse the 800 sentences the reader is scored on; the scope and lifetime phrases were committed with the training templates, not before them; weekday and date phrases are not in the language.
- One training run per adapter (readers 1 and 2, same recipe, differ by 1.5 points, so smaller differences between readers sit inside seed noise); the simulated operator answers a question with the true meaning, so every gain from asking is an upper bound; the decider was trained on the first round's intents, so "decider alone" measures a training mismatch as well as the architecture; combination and conflict rules of the table are our choices.
- The 4-bit laptop build loses 1.7 points on the third-format sets, most of it on composition (clause pairs never seen together 96.5 to 90.2 %, triples 91.2 to 83.8 %); the 70B ran from community GGUF builds; the TAPI object was not sent to a controller.
- The mixed stream's proportions are ours. From its per-kind rates the hybrid stays the best of the three policies for a share of unknown types between about 1 % and 42 % of requests (`docs/decisions.md`, 12 Oct), extrapolated, not measured; the simulated operator answers every question with the truth, so the gain from asking is an upper bound.

7. Reproducibility and cost

`make run` rebuilds every table from `results/` without a GPU and replays a laptop episode of the hybrid; `make console` serves the console on port 8090. Every results file records the command, the simulator commit (`f7c4491`), seeds and loads; every model answer is in `results/cache/`; `docs/decisions.md` logs every deviation, mistake and its cost. Hugging Face Jobs on one A100: 17.3 GPU-hours and about USD 43.18 over four rounds (the fourth: USD 3.14). The submission is the commit tagged `submission` in the repository.

8. Related work

LLMs now operate optical networks in testbeds and field trials: a locally fine-tuned 7B that calls management APIs with a digital twin to pre-verify changes (Sun et al., JOCN 2025), an agent with LLM-native, LLM-centric and rule-centric modes on a 440 km field link (Liu et al., JOCN 2026), and agents that drive controllers through their standard interfaces (TAPI, OpenROADM). We add a compliance metric in closed loop on a stream that includes what the model was never shown, a compositional held-out protocol with instruction-blind baselines, and the cost of asking. Intent lifecycle, conflicts and reports are what the standards ask for (ETSI ZSM 011; TM Forum IG1253, TMF921) and what surveys of intent-based networking find missing (Gharbaoui et al., 2026); trust needs read-back and verification (Jacobs et al., TNSM 2025). Asking by showing the difference follows Clarify (HotNets 2025), which resolves ambiguity in configuration synthesis by differential verification. Our compositional result matches Nandy (2026) and Qiu et al. (2022); the second-reader result matches Hamidieh et al. (2026) on cross-model disagreement.

9. Next steps

A slot for a request's own end points; a second reader of the fourth format; policies with calendar times; the TAPI object sent to a controller (TeraFlowSDN); instructions written by operators; the Swiss network in closed loop.

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